

Seat No: _____

[58]

SARDAR PATEL UNIVERSITY
BCA (3rd SEM) EXAMINATION (NEP) OCT-2025
US03AEBCA06 : DISCRETE MATHEMATICS

No of printed page: 2



Date: 10/10/2025 (Friday)

Time: 02:00PM to 03:30PM

Marks:25

Q-1 Select appropriate answer.

[05]

- 1 $A = (a_{ij})$ is upper triangular matrix if its elements _____ are all zero
(a) above the diagonal (b) below the diagonal
(c) (1) & (2) both (d) none
- 2 Let $u = (0, 4, 2)$, $v = (1, 0, 2)$ then $u \cdot v =$ _____
(a) 4 (b) 9 (c) 0 (d) -4
- 3 The determinant of $\begin{pmatrix} -4 & -2 \\ -2 & 1 \end{pmatrix}$ is _____
(a) 8 (b) -8 (c) 0 (d) 1
- 4 The number of edges in the complete graph K_8 is _____
(a) 28 (b) 27 (c) 29 (d) 0
- 5 An Edge whose endpoints are the same vertex is called _____
(a) trivial graph (b) multi graph (c) loop (d) multiple edges

- Q-2 Let $u = (5, 4, 1, 2, 5)$, $v = (-3, 0, 4, 7, 3)$ and $w = (0, 5, -8, 2, 3)$ Find (i) $3u + (-2)v + 6w$
(ii) Is $(u + v)w = uw + vw$?
(iii) Is $(u + v)w = (v + w)u$?
(iv) $\|u + w\|$
(v) $u \cdot v$
- [10]

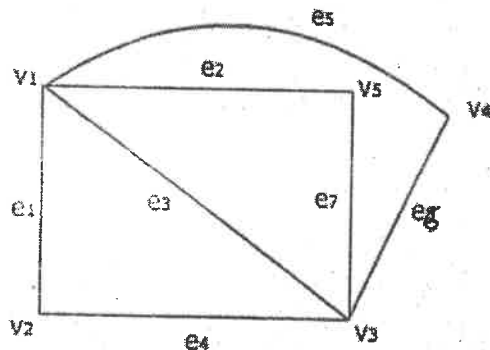
OR

- Q-2 For given matrices, $A = \begin{bmatrix} 2 & 0 & 6 \\ 1 & 2 & 5 \\ 1 & -3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 3 & -1 & 5 \\ 2 & 2 & 6 \\ 7 & 6 & 0 \end{bmatrix}$

- (i) Is $AB = BA$? (ii) Find $\det(A - B)$.

Q-3 For given graph,

[10]

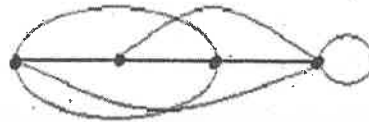
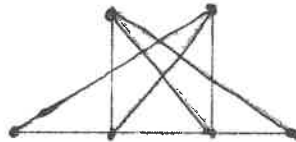


(i) Find the Adjacency matrix and the Incidence matrix .

(ii) Find all possible Simple paths from vertex v_1 to v_3 .

OR

Q-3 Define planer graph. And check which of the following are planar graphs. Justify your answer. Also verify Euler's formula for given graphs. [10]



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